Local Scaling and Temporal Learning in Chart-Based Return Prediction

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Abstract

We study what a chart CNN learns from recent market histories. Chart construction combines local high–low scaling, a binary grayscale image, spatial layout, and flexible learning. We separate these components by comparing the chart CNN with logistic and CNN1D models applied to the same underlying price and volume sequences under alternative normalizations. Local high–low scaling exposes a strong weekly return ranking before nonlinear feature extraction. CNN1D applied to the locally scaled sequence then reproduces the chart CNN's portfolio performance across input windows. Chart and sequence scores rank many of the same stocks, and multimodal fusion adds little to the stronger representation. These findings identify the chart primarily as a delivery mechanism for a locally normalized temporal signal. Transparent price rules, transaction-cost adjustments, weighting schemes, and tradability screens delimit its economic content. The historical performance is concentrated in equal-weighted portfolios and in small, illiquid stocks, becomes negative under modest investability screens, and attenuates during 2020–2024. Chart-based predictability therefore reflects representation and temporal learning, with limited chart-specific and implementable content.

Keywords: Return predictability, Machine learning, Representation learning, Technical analysis, Transaction costs

JEL classification: C45, G11, G12, G14

1. Introduction

Chart-based models raise a basic question for return prediction. When a machine learns from a market chart, what has it learned? The answer shapes how we interpret predictability from recent price and volume data. Before the model sees a chart, the input window has already been selected, locally normalized, and rendered into a visual object. Strong chart CNN performance may therefore reflect the image layout, the local scaling of the input, or the ability of a flexible learner to extract nonlinear structure from price-volume information.

We study the representation question through three learned model families. The chart CNN learns from price and volume images, while CNN1D learns from scaled temporal sequences built from the same underlying data. The multimodal model combines the chart and sequence branches. CNN1D tests how much of the chart CNN signal can be recovered without the image layout, and the multimodal model tests whether the two representations contain complementary predictive information.

The sequence models are evaluated under three scaling schemes. Image scaling preserves the normalization embedded in chart construction, while cumulative-return and devolatized-return scaling provide alternative numerical representations of the same recent market history. We apply both a logistic classifier and CNN1D to these inputs. The logistic model shows how much predictive content is present in each encoding, and CNN1D shows what is added through nonlinear temporal learning.

The multimodal model provides a direct test of representation complementarity. If the chart and sequence branches extract distinct predictive information, combining them should improve the stock ranking beyond either branch alone. If their information mostly overlaps, multimodal gains should be limited. We complement this test with score correlations, encompassing regressions, and incremental explanatory-power tests because similar portfolio performance can arise from different stock rankings.

Our empirical analysis uses daily U.S. equity data from CRSP. Model development is confined to 1993–2000, and we report out-of-sample results for the full 2001–2024 period. We divide the evaluation period into a 2001–2019 baseline sample and a 2020–2024 extension sample. We compare the learned models with momentum (MOM), one-month short-term reversal (STR), one-week short-term reversal (WSTR), and a multi-horizon trend strategy (TREND). All methods use the same universe, timing discipline, portfolio construction rules, weighting schemes, and transaction-cost treatment.

The scaling choice is economically large. In the representative specification with a 20-day input and a five-day return horizon, the logistic classifier produces equal-weighted gross Sharpe ratios of 5.22 under image scaling, −0.41 under cumulative-return scaling, and 2.73 under devolatized-return scaling. Image scaling also produces the strongest logistic results across the 5-day and 60-day input windows. The image-scaled encoding therefore contains substantial predictive content before nonlinear temporal learning is added.

CNN1D models trained on image-scaled temporal sequences often match or outperform the chart CNN. At the five-day return horizon, CNN1D reaches equal-weighted gross Sharpe ratios of 6.24, 6.55, and 5.89 across the 5-day, 20-day, and 60-day input windows. The corresponding chart CNN values are 5.88, 6.52, and 3.72. The weekly signal therefore largely survives the move from chart image to temporal sequence under image scaling. At longer return horizons, performance is weaker and the additional contribution of nonlinear temporal learning is less consistent.

The multimodal evidence points to limited complementarity between the chart and sequence representations. Their scores have Spearman correlations of 0.73 and 0.67 in the two strongest weekly specifications, and their high-minus-low returns have correlations above 0.9 in the representative case. The multimodal model performs well, with a gross Sharpe ratio of 6.48 in the 20-day-input specification, compared with 6.52 for the chart CNN and 6.55 for CNN1D. Its limited improvement indicates that much of the predictive information is already present in the two single-branch models.

The longer-window weekly case shows stronger representation-specific variation. When the 60-day input is used to predict the five-day return, the chart CNN and CNN1D score correlation falls to 0.44. CNN1D also preserves its rankings more consistently across input windows than the chart CNN. The encompassing regressions and incremental explanatory-power tests identify CNN1D as the most consistent source of incremental predictive content, while the chart CNN retains independent content in selected specifications.

We compare the learned models with MOM, STR, WSTR, and TREND under common portfolio construction rules. In the full-sample equal-weighted weekly comparison, WSTR is the strongest gross benchmark with a Sharpe ratio of 2.86, followed by TREND at 2.49. The strongest learned specifications produce gross Sharpe ratios around 6.5. The learned-model advantage is clearest in equal-weighted weekly portfolios and narrows at longer horizons and under value-weighting.

Trading assumptions materially affect the economic interpretation. The representative full-universe equal-weighted weekly portfolios retain net Sharpe ratios near four after the size-dependent 10–20 basis-point transaction-cost adjustment. Value-weighted weekly portfolios are weak or negative after costs. The portfolios remain positive with much weaker performance after a price filter, while dollar-volume, no-microcap, and combined tradability filters produce negative net returns. The economically large signal is therefore concentrated in the smaller and less liquid part of the cross-section.

Performance weakens in the 2020–2024 extension sample. The representative learned models have equal-weighted net Sharpe ratios between 4.78 and 5.28 in the 2001–2019 baseline sample, compared with 1.11 to 1.35 in the extension sample. WSTR and TREND also weaken during 2020–2024. The decline therefore appears across learned models and price-based benchmarks, and the learned-model advantage narrows relative to the baseline sample. Several equal-weighted learned specifications still retain positive net performance.

1.1. Related literature

Our paper relates first to research on technical analysis and return predictability from price histories. Lo et al. (2000) formalize visual chart patterns through systematic statistical recognition. Jiang et al. (2023) replace prespecified patterns with a CNN trained directly on chart images. Murray et al. (2024) use machine learning on numerical histories of monthly returns and find nonlinear predictive content distinct from standard momentum and reversal, including among the largest 500 stocks. The chart CNN replication of Kaczmarek and Pukthuanthong (2025) places durability, microcap concentration, and implementation at the center of the image-based evidence. We study a different margin. Our tests hold recent price and volume histories fixed and ask which transformations of those histories explain the chart model's ranking.

The second connection is to machine learning in empirical asset pricing. Flexible methods can capture nonlinear interactions that linear models miss (Gu et al. 2020), while recent evidence shows that carefully specified regressions can remain competitive in stock-return prediction (Cheng et al. 2025). Our results connect these findings through input representation. Local scaling creates a strong linear ordering, and temporal convolution adds most consistently at the weekly horizon. Model complexity and data construction consequently enter the prediction problem jointly.

The third connection is to the economic evaluation of return predictors. Trading costs can reverse conclusions drawn from gross anomaly returns (Novy-Marx and Velikov 2016), and predictor performance commonly declines outside the discovery sample (McLean and Pontiff 2016). The concentration of weekly reversal among small and illiquid stocks provides a particularly close comparison for our short-horizon portfolios (Chen et al. 2025). We combine these implementation tests with representation diagnostics, so the evidence distinguishes what the models learn from where the resulting portfolio can be traded.

Against this background, we make three contributions. First, we identify local high–low scaling as a concrete source of chart CNN performance and show that a continuous temporal model can reproduce the central weekly portfolio. Second, we connect representation choice to signal substitutability through rank correlations, encompassing regressions, and multimodal fusion. Third, we map the signal's economic scope using transparent price rules, transaction-cost estimates, weighting schemes, liquidity screens, factor regressions, and a post-2019 evaluation. Together, these tests identify the representation mechanism and place strict limits on its tradable scope.

The rest of the paper proceeds as follows. Section 2 describes the data, representations, models, and portfolio design. Section 3 presents the scaling and temporal-learning results and studies overlap across learned signals. Section 4 compares the models with transparent price rules and evaluates costs, tradability, and sample stability. Section 5 concludes. The online appendix reports architecture details, complete horizon grids, auxiliary fusion estimates, and supporting portfolio tables.

2. Data, representations, and empirical design

The empirical design changes the representation of recent market history in a sequence of increasingly flexible prediction models. The first comparison holds the learner simple and varies the normalization. The second holds local normalization fixed and changes the learner and input geometry. The third supplies both chart and sequence representations to a common predictor. We then translate every score into portfolios using the same ranking, holding-period, weighting, and transaction-cost conventions. This organization treats chart construction as the mechanism to be explained and portfolio performance as evidence about its economic content.

2.1. Sample and prediction target

We use daily U.S. equity data from CRSP. The source universe contains ordinary common shares with share codes 10 and 11 listed on NYSE, AMEX, or Nasdaq. The processed panel provides daily returns, open, high, low, and close prices, volume, lagged market capitalization, stock identifiers, and delisting returns when available. Firms enter and leave with their CRSP histories, which produces an unbalanced panel without a survivorship requirement. Positive price fields and the history required by a given signal must be available at the formation date.

The full workflow spans 1993–2024. We use 1993–2000 for model development and reserve 2001–2024 for portfolio evaluation. The evaluation sample is divided into a 2001–2019 baseline period and a 2020–2024 extension period. No observation from either evaluation period enters neural-network fitting or checkpoint selection.

Let $I \in \{5, 20, 60\}$ denote the number of trading days in the input window and $R \in \{5, 20, 60\}$ the future return horizon. We refer to a model with a 20-day input and a 5-day target as $I20/R5$. For stock i at formation date t, the directional label is

$$y_{i,t}^{(R)} = \mathbb{1}\left(\prod_{j=1}^{R}(1 + r_{i,t+j}) - 1 > 0\right). \tag{1}$$

The return on date t is excluded from the target. Every input uses information available by the close of date t.

Raw nominal prices can jump mechanically after splits. We therefore reconstruct a price-consistent close path from daily returns within each lookback window and scale open, high, and low prices relative to that path. The moving-average channel uses a window equal to I. These reconstructed open, high, low, close, moving-average, and volume histories form the common economic input to the chart and sequence models.

History requirements differ across methods. Learned inputs require 5–60 days, the MOM benchmark reaches 273 trading days into the past, and the longest moving-average feature reaches

1,000 days. Standalone portfolios draw from the common CRSP source universe and use common downstream rules, while signal-specific histories can produce different eligible cross-sections. The learned-model overlap tests impose an exact common stock-date panel. We preserve this distinction throughout the paper because portfolio performance and signal overlap answer different questions.

2.2. From market histories to charts

A price chart makes two conceptually separate changes to the underlying observations. It first places the price path on a local scale and then draws the scaled values in a two-dimensional field. The first operation determines which price histories appear similar across stocks. The second determines which local patterns a spatial filter can detect. Our representation tests separate these operations.

For each stock-window, let $m^{(I)}_{i,t}$ and $M^{(I)}_{i,t}$ be the minimum and maximum across all open, high, low, close, and moving-average observations in the window. Each price-channel observation $x_{i,t-j}$ is transformed into

$$\widetilde{x}_{i,t-j} = \frac{x_{i,t-j} - m^{(I)}_{i,t}}{M^{(I)}_{i,t} - m^{(I)}_{i,t}}. \tag{2}$$

Volume is min–max scaled separately within the same window.

The scaled price channels are drawn as daily open-high-low-close bars with a moving-average line. Volume occupies a separate lower panel. The 5-, 20-, and 60-day images have dimensions 32×15, 64×60, and 96×180 pixels. Pixels belonging to a bar, line, or volume element take value one and empty pixels take value zero. The finished chart removes the nominal price level, preserves local geometry, and discretizes the continuous sequence.

The numerical sequence contains the same six channels in chronological order. We study three sequence encodings. The image-scaled sequence retains the continuous values from Eq. (2) and separately scaled volume. The cumulative-return scale anchors each price channel to the first close in the window. The devolatized-return scale expresses price changes relative to lagged exponentially weighted volatility and represents volume through relative changes. Scheme-specific channels are standardized using moments estimated in the development sample and then fixed for the evaluation period.

We first remove mechanical discontinuities in the nominal price record. The first close in each lookback window is anchored to one, subsequent closes are reconstructed from returns, and open, high, and low prices are expressed consistently with that reconstructed path. We then apply the representation-specific normalization. For the chart and image-scaled sequence, the minimum and maximum are computed jointly across the five price channels within the stock-window, while volume is scaled against its own within-window range. Each stock receives a new scale at every formation date using only its own recent history.

The two representations diverge after this common normalization. The sequence model retains the scaled open, high, low, close, moving-average, and volume observations as continuous, chronologically ordered channels. The chart maps the scaled price variables to a fixed-height price panel, reserves a separate lower panel for volume, and renders the result as a binary pixel matrix. The chart consequently discards within-pixel numerical precision while preserving the relative location and shape of the recent path. This ordering isolates the economic role of local scaling more closely than a comparison between raw numerical inputs and chart images. A strong image-scaled sequence model shows that chart-style normalization carries predictive information even when binary rendering and two-dimensional layout are removed.

Table 1 shows how the models turn these encodings into a decomposition. The logistic comparisons measure how much cross-sectional ranking is already present in each encoding. CNN1D adds nonlinear temporal feature extraction to the same numerical sequence. The chart CNN combines local scaling, binary rendering, and two-dimensional convolution. The multimodal model supplies both encodings to paired branches.

Table 1: Models in the representation decomposition

Model	Input scale	Representation	Learner
Logistic, cumulative	Initial-close anchor	Continuous ordered channels	Linear logit
Logistic, devolatized	Lagged volatility	Continuous ordered changes	Linear logit
Logistic, image scale	Local min–max	Continuous ordered channels	Linear logit
CNN1D, image scale	Local min–max	Continuous ordered channels	Temporal convolution
Chart CNN	Local min–max	Binary two-dimensional chart	Spatial convolution
Multimodal	Local min–max	Continuous sequence and binary chart	Paired branches with fusion

Note. All inputs contain reconstructed open, high, low, close, moving-average, and volume histories from the same stock-window. The comparison narrows the source of performance but also changes input precision and convolution geometry when moving between chart and sequence models.

The logistic specifications use the same directional target as CNN1D and map the flattened sequence into a linear index. Comparing them across scaling schemes measures how much ranking information is exposed by normalization alone. Comparing logistic and CNN1D under local scaling measures the increment associated with nonlinear interactions across adjacent observations. The chart CNN/CNN1D comparison then asks whether binary spatial encoding produces an economically different ranking after both models receive the same locally normalized history.

The final comparison studies complementarity. Multimodal inputs are paired by stock, formation date, input interval, and forecast horizon. A stable improvement from fusion would show that the chart and sequence branches extract differences that are useful in the portfolio sort. Similar branch and fusion performance would point toward a common economic signal. This evidence is descriptive because the fusion rule and optimization can constrain the joint model. The central interpretation therefore rests on patterns that recur across scaling schemes, input windows, learners, and the separate overlap tests.

The design narrows the source of performance without creating a pure treatment for layout. Chart and sequence models differ in numerical precision, dimensionality, filter geometry, depth, and capacity. We interpret similar performance after local scaling as evidence that the pixel image is unnecessary for reproducing the tested ranking. Residual differences can arise from any of the remaining representation or architecture features.

2.3. *Prediction models and estimation*

The chart CNN follows the architecture used for chart-based return prediction in Jiang et al. (2023). Each convolutional block applies a 5×3 filter, batch normalization, a leaky rectified linear activation, and max pooling. The 5-, 20-, and 60-day networks contain two, three, and four blocks.

After the final block, the feature map is flattened, dropout is applied, and a linear layer produces two return-sign logits. This architecture establishes the chart benchmark whose performance the numerical models seek to explain.

CNN1D processes the six numerical channels along the time dimension. Its filters span three adjacent trading days and observe every channel at each location. The 5-, 20-, and 60-day variants contain one, two, and three convolutional blocks, with 64 channels in the first block and twice as many after each subsequent block. The output layer matches the chart model's two-class prediction task.

The main multimodal specification pairs the chart and CNN1D branches for the same stock, date, I, and R. If $f_{i,t}^{2D}$ and $f_{i,t}^{1D}$ are the learned branch features, the joint representation is

$$f_{i,t}^{MM} = \left[f_{i,t}^{2D}; f_{i,t}^{1D}\right]. \tag{3}$$

A linear classifier maps the concatenated vector to the return-sign logits. We also examine late fusion through probability or logit averaging. Fusion performance is a suggestive complementarity diagnostic. Optimization, calibration, parameter count, and the training seed can also limit any gain from combining branches.

Each model produces a positive-class probability

$$s_{i,t} = \frac{\exp(z_{i,t}^{(1)})}{\exp(z_{i,t}^{(0)}) + \exp(z_{i,t}^{(1)})}, \tag{4}$$

which is used only as a within-date ranking score. We make no calibration claim for $s_{i,t}$ as an expected return or physical probability.

We estimate the model grid separately by input interval, forecast horizon, model family, and, where applicable, scaling or fusion design. Every neural specification minimizes cross-entropy loss with Adam, a learning rate of 10^{-4}, batches of 128, dropout of 0.50, Xavier initialization, and at most 50 epochs. We retain the checkpoint with the lowest validation loss and stop after two epochs without improvement. Each fitted model solves the same positive versus non-positive return classification problem. The selected checkpoint supplies one score for each eligible stock-date observation in the evaluation period, and model parameters are not updated after 2000.

Development observations from 1993–2000 are assigned to a 70% training set and a 30% validation set using a fixed seed. Because the split is taken over stock-date observations, correlated observations from the same firm or calendar period can appear in both subsets. Validation performance therefore measures fit within the development environment and is not evidence of economic performance in a later period. The economically relevant separation occurs at the 2001 boundary, after which observations enter portfolio evaluation without further fitting or checkpoint selection. A temporally blocked validation design would provide a stricter tuning exercise.

The main empirical analysis uses one selected checkpoint for each learned specification, so a cell can reflect both its representation and the optimization outcome of that fitted network. For the chart CNN, matched five-member forecast averages are also available for $I5/R5$ and $I20/R5$. Those sensitivity results are reported in the online appendix and leave the weekly conclusions essentially unchanged, but they do not constitute a repeated-seed design for every model family and horizon. We consequently place greater weight on patterns that recur across input lengths, model classes, portfolio weighting schemes, and economic diagnostics.

2.4. *Portfolio construction*

Only the ordering of scores within a formation-date cross section enters the sort. The positive-class probability is not converted into an expected return, and differences in probability calibration across model families do not receive a direct portfolio weight. Each eligible cross section is divided into deciles. The high-minus-low strategy buys decile 10 and shorts decile 1. Equal weighting gives each constituent the same weight within its leg. Value weighting uses market capitalization observed before formation and normalizes weights separately within the long and short legs. Each leg carries one dollar of notional, so the spread has zero net investment and two dollars of gross exposure.

The timing convention links the statistical target directly to the traded ranking. At formation date t, every model input ends with information available by the close of that date. The associated score ranks stocks for the return realized from $t + 1$ through $t + R$; the date-t return is outside the target. We retain one formation date per holding interval, producing non-overlapping weekly, monthly, and quarterly return series for $R = 5$, 20, and 60. For a portfolio return series $R_{p,t}$ with $K \in \{52, 12, 4\}$ observations per year, the reported Sharpe ratio is

$$\mathrm{SR}_p = \sqrt{K}\,\frac{\overline{R}_p}{\mathrm{sd}(R_{p,t})}. \tag{5}$$

No risk-free rate is subtracted from the zero-net-investment spread.

Eligibility is signal-specific. A learned model requires its own lookback window, while benchmark signals can require histories extending to 273 or 1,000 trading days. Portfolio comparisons therefore apply common ranking, horizon, weighting, and cost rules to methods drawn from the same CRSP source universe, but they need not contain the exact same stock-date observations. The overlap diagnostics impose the stricter common panel for the three learned scores. This distinction keeps the portfolio exercise representative of each available signal while reserving matched-observation comparisons for questions about ranking overlap and conditional predictive content.

The paper reports complete grids because input length, horizon, weighting, and sample period create meaningful heterogeneity. We use $I20/R5$ as a representative specification in focused tables and economic tests. It is centrally located in the short-horizon grid and performs strongly for each learned family. This choice was made after examining the grid, so best-cell comparisons are descriptive.

2.5. *Price-based benchmarks*

Four transparent price-based benchmarks show how much of the learned performance is attainable with interpretable signals. Momentum (MOM) follows the past-return logic of Jegadeesh and Titman (1993) and uses the cumulative return over 252 trading days after skipping the most recent 21 days,

$$z_{i,t}^{\mathrm{MOM}} = \prod_{j=22}^{273}(1 + r_{i,t-j}) - 1. \tag{6}$$

Excluding the most recent 21 trading days separates medium-horizon momentum from the short-run reversal patterns captured by STR and WSTR.

The one-month and one-week reversal scores follow Jegadeesh (1990) and Lehmann (1990) and are

$$z_{i,t}^{\mathrm{STR}} = -\left(\frac{P_{i,t}}{P_{i,t-21}} - 1\right), \qquad z_{i,t}^{\mathrm{WSTR}} = -\left(\frac{P_{i,t}}{P_{i,t-5}} - 1\right). \tag{7}$$

High scores identify recent losers. The processed price path used for these rules is split-consistent with the return history. WSTR is the closest explicit rule to the learned weekly target.

The final benchmark adapts the multi-horizon trend construction of Han et al. (2016). We use the moving-average lag set

$$\mathcal{L} = \{3, 5, 10, 20, 50, 100, 200, 400, 600, 800, 1000\}$$

and define $x_{i,t,L} = \mathrm{MA}_{i,t,L}/P_{i,t}$ for $L \in \mathcal{L}$. Cross-sectional regressions map these ratios into the next holding-period return. The forecasting coefficients are the trailing mean of the previous 12 fully realized coefficient vectors. The date-t score is

$$z_{i,t}^{\mathrm{TREND}} = \sum_{L \in \mathcal{L}} \overline{\beta}_{L,t} x_{i,t,L}. \tag{8}$$

TREND is a supervised linear benchmark with lagged coefficient estimation. Its transparency comes from explicit features, observable lagged coefficients, and a linear rolling forecast.

2.6. *Cost adjustment and implementation tests*

For leg $\ell \in \{\text{long}, \text{short}\}$, conventional half-turnover is

$$\mathrm{Turnover}_t^{\ell} = \frac{1}{2} \sum_i \left| w_{i,t}^{\ell,\mathrm{target}} - w_{i,t}^{\ell,\mathrm{prev}} \right|. \tag{9}$$

Reported turnover sums the two legs and converts the value to a monthly equivalent. The cost calculation uses the full absolute traded notional, equal to twice the conventional half-turnover measure. We apply a one-way cost of 10 basis points to stocks above the contemporaneous NYSE 80th-percentile market-cap breakpoint and 20 basis points to other stocks. Net return is

$$R_{p,t}^{\mathrm{net}} = R_{p,t}^{\mathrm{gross}} - \sum_{i,\ell} c_{i,t} \left| \Delta w_{i,t}^{\ell} \right|. \tag{10}$$

The schedule provides a common implementation stress test. A capacity assessment would additionally require stock-specific spreads, price impact, borrow fees, locate constraints, and short-sale availability.

We therefore re-form the representative weekly portfolios after imposing more demanding investability screens. The screens require a price of at least \$5, dollar volume of at least \$1 million or \$5 million, market capitalization above the NYSE 20th-percentile breakpoint, or all three strict conditions jointly. The models are not re-estimated. We also regress weekly net high-minus-low returns on the market, size, value, and momentum factors of Carhart (1997) and report annualized intercepts with Newey–West standard errors using four lags.

2.7. *Signal overlap and conditional content*

Similar portfolio returns can conceal different stock selections. For each (I, R) cell, we construct the exact stock-date intersection on which the chart CNN, CNN1D, and multimodal scores and future returns are all observed. We compute date-level Spearman rank correlations (Spearman 1904) and average them through time. We also correlate the resulting high-minus-low returns. The first statistic compares full cross-sectional orderings; the second asks whether differences in those orderings survive in the traded tails.

We also estimate date-level cross-sectional regressions using the Fama–MacBeth procedure (Fama and MacBeth 1973),

$$R_{i,t+1:t+R} = \alpha_t + \beta_t^{2D} s_{i,t}^{2D} + \beta_t^{1D} s_{i,t}^{1D} + \beta_t^{MM} s_{i,t}^{MM} + \varepsilon_{i,t+R}. \tag{11}$$

The Fama–MacBeth coefficients are averages of the date-level slopes. Their conventional test statistic divides the time-series mean by $\mathrm{sd}(\beta_t)/\sqrt{T}$. It does not apply an autocorrelation correction, so the significance levels are exploratory, especially at longer horizons, where the number of formation dates is smaller and persistence in the coefficient series can remain despite non-overlapping portfolio sampling. This convention differs from the weekly Carhart tests (Carhart 1997), for which we use Newey–West standard errors with four lags.

The coefficient magnitudes also require care. The regressors are positive-class probabilities produced by separately trained models, so their numerical scales can differ even when their induced rankings are similar. Correlation among the three scores can make a joint slope sensitive to which other scores enter the regression. We therefore use the encompassing regressions to test conditional association within the exact common stock-date panel. We do not interpret a coefficient as a structural contribution of an architecture or as the return generated by an independent trading signal. The rank correlations and portfolio sorts remain the more direct evidence for whether the models order stocks similarly and whether that ordering is economically consequential.

The wider model grid creates an additional inferential boundary. Input length, forecast horizon, scaling, architecture, fusion, weighting, and sample period generate many related comparisons. The representative $I20/R5$ specification was selected after examining that grid, and we do not apply a formal multiple-testing correction. Claims about the representation mechanism therefore rely on recurring cross-cell patterns and on the separate benchmark, cost, tradability, and subperiod evidence. Raw incremental in-sample R^2 is excluded because adding regressors raises that statistic mechanically. Taken together, these choices make the tests informative about the empirical content and limits of the learned rankings while stopping short of search-adjusted or causal inference.

3. What the chart CNN learns

The chart model's output combines the effect of its input transformation with the effect of convolution. We begin with the transformation. A locally scaled logistic model reveals whether chart construction has already exposed a return ranking before flexible features are learned. We then introduce temporal convolution, compare its stock ordering with the chart CNN, and test whether a joint model benefits from observing both representations.

3.1. Local scaling exposes the weekly signal

Table 2 reports the primary decomposition. Panel A changes the scale while keeping a logistic learner. Panel B compares temporal convolution, spatial convolution, and joint fusion after local scaling. The columns show gross Sharpe ratios over 2001–2024 for three input lengths and both weighting rules. The table focuses on the weekly target, where the learned signals are strongest. Complete monthly and quarterly results appear in the online appendix.

Panel A shows that normalization changes the return ranking before model complexity becomes relevant. With a 20-day input, the equal-weighted logistic Sharpe ratio moves from −0.41 under cumulative-return scaling to 2.73 under devolatized scaling and 5.22 under image scaling. The same ordering appears for 5- and 60-day inputs. Image scaling produces Sharpe ratios of 4.88 and 5.44 in those columns, while cumulative scaling remains near zero or negative. The

Table 2: Local scaling and model architecture in weekly return prediction

	5-day input		20-day input		60-day input	
Model	EW	VW	EW	VW	EW	VW
Panel A. Encoding with a logistic learner						
Logistic, cumulative scale	0.18	−0.04	−0.41	−0.51	−0.26	−0.53
Logistic, devolatized scale	2.80	0.79	2.73	0.48	2.70	0.66
Logistic, local image scale	4.88	1.25	5.22	1.09	5.44	1.12
Panel B. Flexible models with local image scale						
CNN1D	6.24	1.16	6.55	1.12	5.89	2.06
Chart CNN	5.88	1.39	6.52	1.44	3.72	1.18
Multimodal	6.01	1.16	6.48	1.52	5.53	1.75

Note. The table reports annualized gross Sharpe ratios of non-overlapping 5-day high-minus-low decile portfolios from 2001 through 2024. EW and VW denote equal- and value-weighted legs. Local image scale applies the chart's within-window min–max transformation to continuous sequence channels. Chart CNN receives the corresponding binary image. Multimodal is the feature-concatenation model. Each row uses that model's available score cross-section; the rows are not restricted to the exact three-model stock-date intersection used in the overlap tests. The online appendix reports the full scaling and return-horizon grids.

recurrence across input windows is the first indication that the normalization embedded in chart construction carries economic information.

Local scaling changes the cross-sectional object seen by the classifier. A given proportional price movement occupies more of the normalized range inside a quiet window and less when the same window contains a large excursion. Opens, closes, daily ranges, and the moving average are jointly scaled within the stock-window. Their position reveals where the latest price lies relative to recent extremes, how the range is distributed through time, and how short-run movements align across channels. The classifier can use these relations directly. Cumulative-return scaling anchors the path at its first close and leaves range invariance to the learner.

The comparison also clarifies why an image can be an effective input. Chart construction removes nominal price levels and gives similarly shaped paths a common height. These choices reduce variation that is unrelated to the within-window configuration while emphasizing endpoints and extremes. The locally scaled continuous sequence retains the same inductive bias without discarding within-pixel precision. Its strong logistic portfolio shows that much of the apparent visual signal is available as a simple relation among normalized observations.

3.2. What temporal learning adds

Panel B asks whether flexible learning requires the spatial chart. Image-scaled CNN1D reaches equal-weighted Sharpe ratios of 6.24, 6.55, and 5.89 across the three input lengths. The corresponding chart values are 5.88, 6.52, and 3.72. The sequence model matches the chart in the 5- and 20-day cases and performs substantially better when a 60-day window forecasts the next week. The result holds across the weekly input grid and is therefore broader than the representative 20-day cell.

The logistic and CNN1D rows locate two parts of the observed performance. Local scaling creates the baseline ranking. Temporal filters then combine adjacent observations and channels, lifting the 20-day-input Sharpe ratio from 5.22 to 6.55. The chart model reaches 6.52 in the same cell. We do not conduct a formal equality test between these selected ratios, but their economic magnitudes are nearly identical. A two-dimensional image is consequently unnecessary for reproducing the central portfolio result.

The wider horizon grid qualifies the weekly result without overturning it. At the 20-day return horizon, the equal-weighted logistic Sharpe ratios under image scaling are 2.03, 2.37, and 1.59 across the 5-, 20-, and 60-day input windows. The corresponding cumulative-scale values are 0.02, −0.29, and −0.30, while devolatized scaling produces 1.27, 1.02, and 0.98. Image scaling therefore remains the strongest linear encoding at the monthly horizon. The same ordering appears at the 60-day return horizon, where the image-scaled logistic model reaches 0.97, 0.94, and 0.55, compared with 0.37, −0.02, and −0.30 under cumulative scaling.

The image-scaled CNN1D remains predictive beyond the weekly target, although temporal convolution does not uniformly improve on the image-scaled logistic model. At the monthly horizon, CNN1D produces equal-weighted Sharpe ratios of 2.18, 2.15, and 1.74, compared with logistic values of 2.03, 2.37, and 1.59. At the quarterly horizon, the corresponding CNN1D values are 0.99, 0.46, and 0.46, compared with 0.97, 0.94, and 0.55 for logistic. The incremental value of convolution is concentrated at the weekly horizon. Local scaling remains the strongest linear encoding at longer horizons, where the return signal itself is also weaker.

Value-weighting produces a smaller signal. Weekly gross Sharpe ratios range from 1.09 to 1.25 for the locally scaled logistic model and from 1.12 to 2.06 for CNN1D. The chart and multimodal models occupy a similar range. Local scaling therefore remains informative among larger stocks, while the dramatic equal-weighted magnitudes arise in the broader cross-section.

The representative $I20/R5$ deciles show how the equal-weighted spread is formed. For the chart CNN, annualized returns move from −31.9% in the lowest-score decile to 53.0% in the highest. CNN1D moves from −34.6% to 54.5%, and the multimodal model moves from −34.3% to 53.3%. Both portfolio legs therefore contribute materially to the high-minus-low result. Returns generally rise with the score across the middle deciles, so the spread is supported by a broader ordering. The final step is unusually steep. Returns rise from 29.5% to 53.0% between deciles 9 and 10 for the chart CNN, from 29.4% to 54.5% for CNN1D, and from 31.1% to 53.3% for the multimodal model. The models create a broad ordering with additional concentration in the highest-ranked tail.

Value-weighting changes both ends of that profile. In the same $I20/R5$ specification, the lowest-decile returns are −2.7%, −0.2%, and −3.4% for the chart, sequence, and multimodal models. Highest-decile returns are 20.5%, 17.9%, and 20.2%. The corresponding spreads remain positive at 23.2%, 18.0%, and 23.6%, but the middle deciles are less orderly and the negative contribution from the low-score portfolio is much smaller. Equal weighting therefore amplifies tail separation and cross-sectional monotonicity. This difference points toward an important role for smaller stocks, which the explicit tradability screens examine below.

The table establishes economic equivalence in portfolio performance. Architectural equivalence lies outside its scope. Binary rendering discards information from the continuous sequence, two-dimensional filters impose different neighborhoods, and the chart model has a different depth schedule. These differences may influence individual scores. The aggregate result is that local scaling and temporal learning are sufficient to reproduce the main weekly portfolio performance.

3.3. Do the models rank the same stocks?

Similar long–short returns can arise from different stock selections. Table 3 reports average date-level Spearman correlations between learned-model scores on their common stock-date panel. The correlations are highest in the weekly cells where portfolio performance is strongest, which links agreement in rankings to the economic result.

Table 3: Rank overlap across learned models and input windows

Specification	Chart/CNN1D	Chart/Multimodal	CNN1D/Multimodal
Panel A. Across-model rank overlap			
$I5/R5$	0.73	0.74	0.84
$I20/R5$	0.67	0.72	0.80
$I60/R5$	0.44	0.65	0.69
$I5/R20$	0.50	0.67	0.71
$I20/R20$	0.53	0.65	0.72
$I60/R20$	0.44	0.55	0.63
$I5/R60$	0.45	0.45	0.68
$I20/R60$	0.44	0.45	0.64
$I60/R60$	0.38	0.46	0.63
Panel B. Within-model rank stability for the 5-day return horizon			
Model	$I5/I20$	$I5/I60$	$I20/I60$
Chart CNN	0.60	0.22	0.34
CNN1D	0.64	0.54	0.71
Multimodal	0.56	0.44	0.59

Note. Panel A reports time-series averages of formation-date Spearman correlations on the common stock-date panel for which all three learned scores and the future return are observed. Panel B compares scores from two input windows within the same model family. All statistics use the 2001–2024 evaluation window.

For $I5/R5$, the chart CNN/CNN1D correlation is 0.73, the chart CNN/multimodal correlation is 0.74, and the CNN1D/multimodal correlation is 0.84. The corresponding values for $I20/R5$ are 0.67, 0.72, and 0.80. The models therefore produce substantially overlapping cross-sectional orderings while leaving room for representation-specific selections.

The portfolio payoffs overlap even more. In the representative $I20/R5$ cell, pairwise correlations between the weekly high-minus-low returns range from 0.92 to 0.94. The return correlations show that differences in the full stock ordering translate into much smaller differences in the traded high-minus-low payoffs. The rising decile profiles in the online appendix confirm that the economic ordering extends beyond one unusual endpoint portfolio.

The 60-day input is the clearest exception. The chart CNN/CNN1D score correlation falls to 0.44 for the weekly target. The chart Sharpe ratio also falls to 3.72, while CNN1D retains 5.89. Panel B shows that the chart's weekly rank correlations involving the 60-day input are only 0.22 and 0.34, compared with 0.54 and 0.71 for CNN1D. Both encodings use the same local range and preserve chronological location, so this divergence cannot be assigned to timing information. Differences in numerical precision, chart construction, filter geometry, depth, and capacity all remain plausible sources. The result shows that representation substitution is weaker in this cell.

Joint cross-sectional regressions provide a conditional version of the same evidence. Table 4 reports conventional Fama–MacBeth t-statistics from regressions using the three raw positive-class probabilities in Eq. (11). In the weekly 5- and 20-day input cells, every score is positive after conditioning on the other learned scores. In the 60-day-input weekly cell, the chart statistic becomes weak while the sequence statistic remains large. At monthly and quarterly horizons, CNN1D is the most consistently positive score.

The conditional evidence is selective. The chart statistic is positive in the $I5/R5$, $I20/R5$, and $I20/R20$ cells and weak elsewhere, while CNN1D is positive throughout the grid. The multimodal score is strongest at weekly horizons. Average date-level joint-regression R^2 ranges from 0.50% to

Table 4: Joint encompassing regressions for learned-model scores

Specification	Chart CNN	CNN1D	Multimodal	Average R^2 (%)
$I5/R5$	6.95	12.56	8.39	0.70
$I20/R5$	9.51	11.80	11.91	0.80
$I60/R5$	−1.12	18.19	9.95	0.95
$I5/R20$	−0.26	4.39	−0.73	0.52
$I20/R20$	3.93	5.76	3.36	0.65
$I60/R20$	−0.85	4.66	0.28	1.06
$I5/R60$	−0.74	2.16	2.56	0.50
$I20/R60$	−0.50	2.83	1.37	0.75
$I60/R60$	−0.46	3.97	0.71	1.13

Note. The first three columns report the conventional Fama–MacBeth t-statistic for each score in a joint date-level cross-sectional regression over 2001–2024. The statistic uses the time-series standard error $\mathrm{sd}(\beta_t)/\sqrt{T}$ without an autocorrelation correction. It is therefore exploratory, especially at longer horizons and in view of the model search. The last column is the time-series average of the date-level joint-regression R^2.

1.13%. These estimates show residual conditional score associations, especially for the sequence model. Their conventional standard errors, model search, and probability-scale differences make them exploratory. The encompassing regressions therefore complement the rank and payoff correlations without establishing independent return mechanisms or isolating an architectural effect.

3.4. Can fusion improve the common signal?

The rank correlations leave meaningful disagreement between the chart and sequence scores. Fusion tests whether that disagreement improves a single predictor. In Table 2, the multimodal model's equal-weighted weekly Sharpe ratios are 6.01, 6.48, and 5.53. Each lies below the better individual branch. The 20-day-input result is especially tight. The chart, sequence, and multimodal Sharpe ratios are 6.52, 6.55, and 6.48.

Late fusion leads to the same conclusion using a distinct estimator. Averaging the pretrained chart and CNN1D probabilities or logits produces full-sample equal-weighted Sharpe ratios of 6.25 for $I5/R5$ and 6.51 for $I20/R5$. These values remain close to the individual branches and do not supply a uniform gain over them. The online appendix reports the late-fusion comparison on the aligned branch predictions.

The fusion evidence is descriptive, and the single-run comparisons do not measure optimization uncertainty. High branch correlations, nearly identical portfolio returns, and limited fusion gains nevertheless support substitutability at the portfolio level. The joint regressions show residual score content, and the rank correlations remain below one. A fused learner can still fail to improve if the residual differences do not materially alter the traded portfolios, are unstable through time, or are difficult to estimate from one training run. The absence of a stable fusion gain therefore supports economic substitutability without establishing informational identity.

4. Economic performance and its limits

The representation tests identify a locally scaled short-horizon ranking shared by several learners. This section evaluates its economic content. We first compare the learned portfolios with explicit price rules. We then examine the transaction-cost adjustment, the part of the cross-section that supports the returns, and the stability of the evidence after 2019.

4.1. Comparison with transparent price rules

Table 5 compares the representative $I20/R5$ learned portfolios with the four price-based benchmarks over 2001–2024. The learned models use the 20-day input; benchmark signals have their own stated lookbacks. Every row uses weekly non-overlapping portfolios and the same weighting, decile, and cost conventions after signal construction.

Table 5: Weekly equal-weighted learned models and price-based benchmarks

Signal	Gross return	Gross SR	Net return	Net SR	Turnover
Chart CNN $I20/R5$	84.9%	6.52	51.3%	3.94	6.64
CNN1D $I20/R5$	89.1%	6.55	56.5%	4.16	6.46
Multimodal $I20/R5$	87.5%	6.48	54.5%	4.04	6.56
MOM	−2.3%	−0.08	−7.1%	−0.25	0.95
STR	44.3%	1.83	27.3%	1.13	3.32
WSTR	59.3%	2.86	24.9%	1.20	6.71
TREND	52.7%	2.49	27.5%	1.31	4.95

Note. Returns and Sharpe ratios are annualized. Turnover is monthly-equivalent conventional half-turnover summed across the two legs. The transaction-cost deduction uses full absolute traded notional. Learned and benchmark strategies draw from the same CRSP source universe, though method-specific lookback requirements can change eligible stock counts.

One-week reversal is the strongest gross benchmark, with a Sharpe ratio of 2.86. TREND reaches 2.49 and one-month reversal reaches 1.83. These substantial returns confirm that recent price histories support strong explicit rankings in the same sample. The learned Sharpe ratios near 6.5 show that local scaling and flexible convolution produce a sharper standalone ordering than any individual rule.

Costs reduce every active weekly strategy. The learned net returns fall by 33–35 percentage points, and the weekly-reversal return falls by 34 percentage points. WSTR and the learned models have almost identical reported half-turnover, near 6.7 times portfolio value per month. Full absolute traded notional is twice this statistic. The learned gross spread is large enough to leave net Sharpe ratios near four under the 10–20 basis-point schedule. WSTR falls to 1.20, while TREND becomes the strongest benchmark at 1.31.

Value-weighting changes the weekly result. The $I20/R5$ chart, CNN1D, and multimodal gross Sharpe ratios fall to 1.44, 1.12, and 1.52. Their net returns are −2.5%, −7.3%, and −1.5%. The full-sample learned edge therefore depends on equal weighting even before more demanding liquidity screens are introduced.

The comparison places the learned results against transparent price-history hurdles. Simple reversal and trend rules are profitable in the same broad short-horizon environment, while the learned models produce larger standalone sorts. The comparison does not measure signal overlap or the fraction of the learned score explained by any benchmark. It also does not establish a new scalable anomaly, because the common cost schedule omits impact, borrowing, and capacity and because the eligible samples are not exact benchmark-by-model intersections.

The learned advantage also contracts as the holding period increases. At $I20/R20$, equal-weighted gross Sharpe ratios are 2.26, 2.15, and 2.20 for the chart CNN, CNN1D, and multimodal model. WSTR reaches 1.12, TREND 0.68, STR 0.59, and MOM 0.06. At $I20/R60$, the learned values fall to 0.34, 0.46, and 0.58. WSTR reaches 0.39 and STR reaches 0.28 in the same quarterly comparison. The learned models remain competitive, but the separation seen in weekly portfolios is largely absent.

Value-weighting compresses the differences further. At $I20/R20$, the three learned Sharpe ratios are 0.49, 0.41, and 0.32, compared with 0.20 for MOM and 0.16 for WSTR. At $I20/R60$, learned-model values of 0.21, 0.15, and 0.01 sit close to MOM at 0.16, STR at 0.15, and TREND at 0.13. Flexible learning therefore adds its clearest economic value in frequent, equal-weighted sorting. It provides little uniform advantage once the target is longer or larger firms receive more weight.

4.2. Where does the portfolio performance reside?

The equal- and value-weighted results suggest that firm size is central to the portfolio evidence. Table 6 locates that dependence more directly by re-forming the representative weekly portfolios after investability screens. Model scores remain unchanged. Each filter is imposed before the decile breakpoints are computed, so the two legs remain the extreme deciles of the eligible cross-section.

Table 6: Tradability and factor diagnostics for representative weekly portfolios

Model	Screen	Net return	Net SR	Carhart α	$t(\alpha)$	Names	Turnover
Chart CNN	Full universe	51.3%	3.94	49.3%	11.81	420	6.65
	Price ≥ \$5	10.1%	1.08	8.1%	3.67	321	6.66
	Dollar volume ≥ \$1m	−10.0%	−0.85	−11.0%	−3.87	241	7.01
	Dollar volume ≥ \$5m	−19.5%	−1.57	−20.6%	−7.27	164	7.07
	No NYSE microcaps	−17.0%	−1.60	−18.1%	−7.38	193	6.89
	Combined	−22.9%	−2.10	−24.0%	−10.21	149	7.02
CNN1D	Full universe	56.5%	4.16	55.2%	12.56	419	6.46
	Price ≥ \$5	13.7%	1.48	12.4%	5.46	319	6.50
	Dollar volume ≥ \$1m	−5.7%	−0.46	−6.2%	−2.21	240	6.68
	Dollar volume ≥ \$5m	−15.0%	−1.17	−15.7%	−5.61	164	6.75
	No NYSE microcaps	−14.8%	−1.37	−15.6%	−6.34	192	6.57
	Combined	−19.0%	−1.68	−19.8%	−8.11	149	6.69
Multimodal	Full universe	54.5%	4.04	53.2%	12.10	419	6.56
	Price ≥ \$5	13.5%	1.52	12.4%	5.46	319	6.63
	Dollar volume ≥ \$1m	−5.2%	−0.42	−5.5%	−1.91	240	6.83
	Dollar volume ≥ \$5m	−14.1%	−1.11	−14.7%	−5.12	164	6.92
	No NYSE microcaps	−14.0%	−1.33	−14.6%	−5.85	192	6.75
	Combined	−19.3%	−1.76	−19.9%	−8.32	149	6.87

Note. The table reports equal-weighted $I20/R5$ portfolios from 2001 through 2024. Names is the average number of stocks in each leg. Combined requires price of at least \$5, dollar volume of at least \$5 million, and market capitalization above the NYSE 20th-percentile breakpoint. Carhart intercepts are annualized and use Newey–West standard errors with four lags.

The gross-to-net change helps separate forecast strength from implementation. In the representative equal-weighted $I20/R5$ portfolios, the transaction-cost adjustment reduces annualized returns from 84.9% to 51.3% for the chart CNN, from 89.1% to 56.5% for CNN1D, and from 87.5% to 54.5% for the multimodal model. Their reported monthly-equivalent half-turnover lies between 6.46 and 6.64. WSTR has comparable turnover at 6.71 and loses 34.4 percentage points of annualized return after costs. The common cost schedule consequently imposes a large penalty on any weekly signal whose ranking changes rapidly.

The remaining equal-weighted net performance does not establish scalability. The cost model applies 10 or 20 basis points to full absolute traded notional, but it omits market impact, borrowing costs, short-sale constraints, and capacity. This distinction already matters under value-weighting. The representative value-weighted weekly net returns are −2.5%, −7.3%, and −1.5% for the three learned models, before the explicit investability filters are imposed.

The price screen removes most of the unrestricted magnitude but leaves positive portfolios. Average leg size declines by roughly one quarter, and net Sharpe ratios fall from about four to 1.08–1.52. Dollar volume and market capitalization are more decisive. A $1 million dollar-volume requirement makes all three net returns negative. Tightening the threshold to $5 million produces net returns from −19.5% to −14.1%, and excluding NYSE microcaps produces net Sharpe ratios from −1.60 to −1.33. The combined screen leaves about 149 stocks in each leg and net returns between −22.9% and −19.0%, with net Sharpe ratios from −2.10 to −1.68. Turnover remains high after every screen, so a smaller eligible set does not produce a more persistent portfolio.

Factor adjustment and tradability therefore answer different questions. Full-universe Carhart alphas remain close to raw net returns, with statistics above 11, so the selected linear factors leave the unrestricted spread intact. The same alphas become negative under the dollar-volume, microcap, and combined screens. This reversal is stronger evidence of illiquid-tail concentration than the equal- versus value-weighted comparison alone. The investable universe sets the main economic boundary, while conventional factor exposure explains little of the unrestricted spread.

This boundary is economically consistent with the short-horizon benchmark evidence. Weekly reversal is associated with temporary price pressure, bid-ask effects, and retail trading, and recent evidence places its largest profits among small and illiquid stocks (Chen et al. 2025). Local scaling and temporal convolution appear to sharpen a related region of the return distribution. The resulting forecast may remain statistically interesting even where the long–short portfolio lacks scalable economic value.

4.3. Does the signal persist after 2019?

Table 7 compares the representative weekly learned models and price-based benchmarks across the baseline and extension periods. The table keeps each signal definition fixed. The five-year extension is much shorter than the 19-year baseline and should be interpreted with correspondingly greater sampling uncertainty.

Table 7: Weekly equal-weighted performance in the baseline and extension periods

	2001–2019				2020–2024			
Signal	Gross ret.	Gross SR	Net ret.	Net SR	Gross ret.	Gross SR	Net ret.	Net SR
Chart CNN $I20/R5$	96.3%	7.64	60.0%	4.78	50.7%	3.87	18.5%	1.35
CNN1D $I20/R5$	102.2%	8.09	66.5%	5.28	49.9%	3.34	18.6%	1.18
Multimodal $I20/R5$	97.6%	7.76	64.4%	5.12	49.5%	3.19	17.2%	1.11
MOM	−5.1%	−0.19	−10.0%	−0.36	8.7%	0.26	4.1%	0.12
STR	48.7%	2.04	31.6%	1.33	27.7%	1.10	11.0%	0.44
WSTR	66.2%	3.33	31.7%	1.59	32.9%	1.42	−0.8%	−0.03
TREND	64.4%	3.09	38.3%	1.85	8.1%	0.38	−13.3%	−0.62

Note. Returns and Sharpe ratios are annualized. The learned rows use the representative 20-day input. Benchmark rows use the signal definitions in Section 2.5. Portfolio and cost conventions are fixed across periods.

Every learned model weakens sharply after 2019. Net Sharpe ratios fall from 4.78 to 1.35 for the chart CNN, from 5.28 to 1.18 for CNN1D, and from 5.12 to 1.11 for the multimodal model. Gross annualized returns fall from 96.3%, 102.2%, and 97.6% to 50.7%, 49.9%, and 49.5%. All three architectures move together to a lower performance region, which is consistent with the shared-signal interpretation. Net returns remain positive at 17.2% to 18.6%, but the fixed transaction-cost schedule consumes a much larger share of the lower gross spread.

The benchmark change extends beyond a neural architecture. WSTR's net Sharpe ratio falls from 1.59 to −0.03, TREND falls from 1.85 to −0.62, and STR falls from 1.33 to 0.44. MOM moves in the other direction, from −0.36 to 0.12. The extension period therefore changes the ranking of both learned and transparent price signals. This co-movement is consistent with a broader shift in short-horizon predictability, but the period split cannot distinguish market adaptation from sampling variation or specification search.

The same learned strategies have negative value-weighted weekly returns after costs, and the full-sample tradability screens show that the unrestricted spread is concentrated in difficult names. The extension evidence therefore supports persistence of a narrow statistical ranking with much weaker economic scope.

Two mechanisms can generate this attenuation. The underlying short-horizon relation may have weakened as market structure, competition, or investor behavior changed. The original estimates can also contain specification search and sampling luck. Our design cannot separate these explanations. A direct chart CNN replication documents deterioration before and after publication (Kaczmarek and Pukthuanthong 2025), while the broader return-predictability literature attributes out-of-sample and post-publication declines to both statistical bias and investor learning (McLean and Pontiff 2016). Locally scaled temporal signals remain detectable after 2019, while their magnitude and implementability are far below the early-sample result.

Annual chart CNN results add a final qualification. The recent deterioration is concentrated in 2020 and 2021, followed by partial recovery during 2022–2024. The five-year extension is also much shorter than the 19-year baseline. The evidence supports attenuation and weaker implementation scope, while a claim of complete disappearance would exceed the reported results. This later-sample evidence is most useful as an economic boundary on the mechanism result. Local scaling and temporal learning continue to produce a detectable ranking, but with much smaller and less stable rewards.

4.4. Economic interpretation

Taken together, the results describe one short-horizon signal seen through several representations. Local scaling exposes the strongest linear ranking. Temporal convolution improves that ranking at the weekly horizon and reaches the chart CNN's portfolio performance. The stock scores overlap, the extreme portfolios move together, and fusion supplies no durable gain. This sequence is consistent with a locally normalized temporal component that both neural architectures recover.

The portfolio evidence also suggests what this component captures. Local scaling jointly normalizes the latest price, recent extremes, intraday ranges, moving average, and volume within each stock-window. A flexible learner can condition the return forecast on the configuration of those variables without carrying information about the nominal price level. The resulting ranking is strongest over the next week and among small, illiquid stocks. Its economic location resembles the part of the cross-section in which temporary price pressure and weekly reversal are strongest (Chen et al. 2025). The reported tests do not establish that the learned score is a nonlinear reversal strategy, since we do not regress it directly on the benchmark signals. They show that the model extracts its largest returns in a related market segment.

This interpretation separates statistical content from investable value. Full-universe factor regressions leave the learned spreads intact. Liquidity screens reverse them. Conventional risk adjustment and tradability therefore deliver different assessments of the same portfolio. The transaction-cost schedule also leaves out market impact, borrowing costs, short-sale constraints, and capacity, all of which are likely to be most severe where the signal is strongest. The negative results under modest dollar-volume screens consequently receive greater economic weight than the large full-universe alphas.

The evidence has further statistical boundaries. Chart and sequence inputs differ in precision and architecture. Neural-network estimates use limited seed variation, standalone portfolios can contain different eligible stock-date observations, and the representative specification was selected after examining a larger grid. The Fama–MacBeth statistics use conventional standard errors and are exploratory. These features favor conclusions that recur across specifications over inference from a single cell. The recurring result concerns the role of local scaling and temporal learning. Residual chart-specific associations, exact performance differences, and post-2019 attenuation require a more conservative interpretation.

The contribution is therefore explanatory. The decomposition shows how a chart CNN can generate striking historical stock sorts from a transformation that is available to numerical models. The economic tests show why those sorts should not be read as evidence of a readily scalable anomaly. Both statements are needed to characterize what the model has learned.

5. Conclusion

Chart-based return prediction combines an economic input transformation with a flexible learner. The familiar image conceals the first part of this combination. Before the network observes recent market history, chart construction removes nominal price levels, scales each stock-window to its own high–low range, aligns price and volume channels, and converts continuous values into a spatial object. We separate these operations to determine what accounts for the resulting return ranking.

The central finding is that the local scaling embedded in chart construction carries much of the signal. A linear classifier applied to conventionally scaled sequences produces little weekly portfolio performance. The same classifier becomes strongly predictive when the continuous sequence receives the chart's within-window scale. CNN1D then recovers the chart CNN's weekly portfolio performance from that locally scaled sequence across input windows. Binary rendering and two-dimensional filters are therefore unnecessary for reproducing the central empirical result. In this setting, the image serves principally as a delivery mechanism for local normalization and temporal learning.

Local scaling changes the prediction problem in an economically meaningful way. It expresses each observation relative to the stock's recent range and jointly scales opens, closes, daily ranges, and moving averages within the stock-window. The learner can compare the location and evolution of these variables across stocks without estimating invariance to nominal price and local volatility from the development sample. Temporal filters add value by combining adjacent observations and channels within this representation. Their incremental contribution is clearest for weekly forecasts, while the normalization effect persists at longer horizons.

The agreement across models supports this interpretation. Chart and sequence scores rank many of the same stocks, and their long–short portfolio returns move closely together. Residual differences appear in the score-level tests, especially for selected windows, yet feature concatenation and late fusion do not generate a stable improvement over the stronger branch. The encodings can influence individual predictions while remaining substitutes for the economically important portfolio ranking. This distinction helps reconcile architectural differences with similar aggregate returns.

The benchmark and implementation exercises define the scope of the mechanism. Transparent reversal and trend rules confirm that the sample contains broad short-horizon price predictability. Learned models create a sharper standalone weekly ranking, although their advantage narrows with longer holding periods and value weighting. Transaction costs absorb a substantial share of the gross returns. More importantly, the net performance is concentrated in equal-weighted portfolios and in the small, illiquid tail of the cross-section. Modest dollar-volume and microcap screens eliminate it. Large unrestricted-sample alphas consequently provide little evidence of a scalable trading opportunity.

Performance also attenuates after 2019 for chart, sequence, and multimodal models, alongside the transparent short-horizon price rules. This shared decline is consistent with a change in the underlying return environment and supplies a further boundary on the historical evidence. The short extension period limits a precise explanation of the change, so the time-series result serves as corroborating evidence for economic fragility rather than as the paper's organizing fact.

Our results place representation design inside the economic interpretation of machine-learning evidence. A model trained on an engineered object can appear to discover a distinctive pattern even when a consequential part of the result was created by the transformation that produced the object. Holding the underlying observations fixed and varying scale, geometry, and learner reveals where predictive content enters the pipeline. For chart-based return prediction, local normalization exposes the ranking and temporal learning refines it; chart-specific content has limited incremental portfolio value.

Further work can sharpen this decomposition. Temporally blocked validation and repeated estimation seeds would better separate representation effects from optimization variation. Direct spanning tests on exact model-benchmark intersections could identify how much of the locally normalized ranking is captured by reversal and trend signals. Stock-specific spreads, market impact, borrowing costs, and capacity constraints would provide a stricter implementation standard. More generally, decomposing other engineered financial representations may reveal whether their apparent economic content originates in the raw history, the transformation, the learner, or their interaction.

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